

Major Goal 2: Graph-First Discovery & Entity Resolution (The Inference Engine)

Objective: Replace direct column-to-column comparisons with isolated sequential extraction followed by graph-level Entity Resolution (ER), safely supporting massive, heterogeneous folders (JSON, PDFs, tabular).

The Right-Way Pipeline (4 Deterministic Steps)

Step 1: Isolated Ingestion & Normalization (Data-Juicer)

When a user targets a folder `/sales_and_billing/`, the engine does **not** cross-compare files.

- **Task:** Data-Juicer processes each file completely independently.
- **Action:** It chunks the files into discrete, chronological Episodes and normalizes the bytes into standard text.

Step 2: Temporal Fact Extraction (Graphiti)

- **Task:** Pass each independent Episode through the Graphiti engine.
- **Action:** Using a lightweight local model (or constrained LLM under budget), Graphiti extracts semantic Entities (e.g., "Justin Mason", "Server-A") and Facts (e.g., "Justin Mason resides in Pune") from the text.
- **Result:** The system builds an isolated graph of entities and facts for that specific file, attaching strict lineage back to the raw block.

Step 3: Cross-System Entity Resolution (ER) Engine

This is where the magic happens and relationships are discovered, without $O(N^2)$ column comparisons.

- **Task:** The ER engine sweeps across the newly populated graph nodes using a tiered approach:
 1. **Blocking:** Uses cheap, deterministic keys (e.g., email domain, tax-id hash, normalized name n-grams) to group potentially related entities.
 2. **Pairwise Scoring:** Only runs semantic/LLM evaluation on entities within the same block to see if `cust_id: 123` in System A is the same person as `client_num: 123` in System B.
 3. **Clustering:** Generates CandidateEdges (Merge Candidates) across systems.

Step 4: The Curation Canvas (Human-in-the-Loop)

- **Task:** The UI does not show a spiderweb of files. It presents the **ER Merge Candidates** and **Conflicts** to the user.
- **Action:** The user sees a cluster: "We found 'Justin Mason' in the CRM and 'J. Mason' in the Billing logs. Do you want to merge these entities?"
- **Result:** When the user clicks [**ACCEPT**], the system writes a `SAME_ENTITY_AS` edge to the Ontology Registry and merges the nodes via a stable ID redirect.

Why This is the Only Way That Works at Scale:

1. **It Supports Everything:** It works flawlessly whether the folder contains 100 spreadsheets or 1,000 PDF contracts, because **Graphiti** extracts entities from text blocks regardless of structure.
2. **Token Cost is Linear, Not Exponential:** You only pay to extract entities from a file *once* ($O(N)$). The cross-comparison is handled cheaply by the ER blocking engine.
3. **Discrepancy Integrity:** If the CRM says a customer is "Active" but the billing log says "Suspended", extracting facts independently allows the system to flag this as a **Conflict** to the user, rather than silently overwriting it during a forced column merge.